Experiment ‘p_3_8_two_pairs’ Results

June 3, 2026

Experiment outcome: SUCCESS
Bad responses: 0
Responses containing `assume` : 0
Responses containing `expect` : 0
Resolution attempts: 1
Hard fails (resolution): 0
Soft fails (resolution): 0
Verification attempts: 1

Problem Specification

Problem name: p_3_8_two_pairs
Natural language statement: Write a method that takes four integers and returns ”two pairs” if the input consists of two matching pairs (in some order) and ”not two pairs” otherwise.
Method signature: p_3_8_two_pairs (i: int, j: int, k: int, l: int) returns (s: string)

Ensures

- `if (i == j && k == l) || (i == k && j == l) || (i == l && j == k) then s == "two pairs" else s == "not two pairs"`

GenAI interactions

Below you will find all interactions between the 'user' (program) and the 'assistant' (OpenAI).

Program → GenAI

```
You are given the following task to perform in Dafny:

Write a method that takes four integers and returns "two pairs" if the input consists of
    ↪ two matching pairs (in some order) and "not two pairs" otherwise.

The signature should be:

p_3_8_two_pairs (i: int, j: int, k: int, l: int) returns (s: string)

The method should respect the following contract:

ensures if (i == j && k == l) || (i == k && j == l) || (i == l && j == k) then s == "two
    ↪ pairs" else s == "not two pairs"

Produce and show only the Dafny body of this method, including the curly braces that
    ↪ surround it. Do not show the signature nor contract. You must not use 'assume' nor
    ↪  'expect' anywhere in your code.
```

GenAI → Program

System fingerprint: null
ID: chatcmpl-Dmdqjs5XMU5UGNziXag7sLewXPICi
Model: gpt-5.5-2026-04-23
Created at: 1780485297
Finish reason: stop
Usage: 398 tokens in, and 191 tokens out

```
{
  if (i == j && k == l) || (i == k && j == l) || (i == l && j == k) {
    s := "two pairs";
  } else {
    s := "not two pairs";
  }
}
```

Final Program

```
method p_3_8_two_pairs (i: int, j: int, k: int, l: int) returns (s: string)
       ensures if (i == j && k == l) || (i == k && j == l) || (i == l && j == k) then s
           ↪ == "two pairs" else s == "not two pairs"
{
  if (i == j && k == l) || (i == k && j == l) || (i == l && j == k) {
    s := "two pairs";
  } else {
```

```
    s := "not two pairs";
  }
}
```

Total Token Usage

Input tokens: 398
Output tokens: 191
Reasoning tokens: 111
Sum of 'total tokens': 589

Experiment Timings

Overall Experiment started at 1780485297267, ended at 1780485308594, lasting 11327ms (11.33 seconds)
Iteration #1 started at 1780485297268, ended at 1780485308594, lasting 11326ms (11.33 seconds)